

PIXELFORENSICS

Deepfake Analysis Report

Report ID: 3851366B958F
Generated: April 24, 2026 at 12:49 PM
Analyzed File: fake_img_16.jpg
File Type: IMAGE



■■ AI-GENERATED / FAKE

Confidence Score: 95.0%

Executive Summary:

Our analysis indicates with 95.0% confidence that this image is likely AI-generated or manipulated. 4 out of 6 detection methods flagged suspicious characteristics. Key indicators include patterns typical of AI generators and anomalies in the image structure that deviate from authentic photographic properties.

Detailed Analysis

Multi-Method Ensemble Detection

Detection Method	Verdict	Confidence	Weight
Secondary detector (98.2% acc)	■ FAKE	69.1%	1.0
Frequency Analysis	■ REAL	70.0%	0.6
ELA Analysis	■ FAKE	95.0%	0.6
Noise Analysis	■ FAKE	95.0%	0.6
Color Analysis	■ REAL	74.0%	0.5
Metadata Analysis	■ FAKE	75.0%	0.8-1.2*

Methods Voting FAKE:	4 / 6
Weighted Ensemble Score:	75.0%
Decision Threshold:	52%

* Metadata weight increases to 1.2 when no EXIF data is found

EXIF Metadata Analysis

■■ NO EXIF METADATA FOUND

What this means: The absence of EXIF metadata is highly suspicious and commonly observed in AI-generated images. However, it can also occur in:

- Screenshots of genuine images
- Images downloaded from websites (metadata often stripped)
- Photos edited in apps that remove EXIF data
- Images shared through messaging apps
- AI-generated or synthetic images (most common)

Technical Details

AI Models Deployed

Model Name	Type	Accuracy	Status
dima806/ai_vs_real	ViT Transformer	98.2%	✓ Active
Ateeqq/ai-vs-human	SigLIP2	99.2%	■ Optional

Detection Methods Explained

Frequency Analysis: Examines the distribution of frequency components in the image using FFT. AI-generated images often show unnatural frequency patterns.

Error Level Analysis (ELA): Compares compression artifacts between original and recompressed versions. AI images show unusually uniform compression patterns.

Noise Analysis: Analyzes sensor noise patterns. Real cameras have characteristic noise signatures that AI generators cannot perfectly replicate.

Color Analysis: Examines color distribution, saturation, and histogram smoothness. AI images often display statistically anomalous color properties.

Metadata Analysis: Checks for authentic camera EXIF data. Real photos contain detailed camera settings, timestamps, and device information.

Conclusion

Based on comprehensive multi-method analysis, we have **very high confidence** that this image is AI-generated or significantly manipulated. Multiple independent detection methods concur on this assessment, exhibiting strong statistical indicators of synthetic generation.

Important Disclaimer

- This report is generated by an automated AI detection system and should be used as a preliminary assessment tool only.
- The results are based on statistical analysis and machine learning models that, while highly accurate, are not infallible.
- For critical decisions (legal proceedings, forensic investigations, or high-stakes verification), this report should be supplemented with expert human analysis and additional forensic tools.
- PixelForensics makes no warranties about the absolute accuracy of these results and accepts no liability for decisions made based on this report alone.
- Accuracy metrics are based on validation testing and may vary with real-world images. The system is regularly updated to detect new AI generation techniques.